Comput Optim Applic (2007) 36:109–133
DOI 10.1007/s10589-006-0393-7

Adjoint concepts for the optimal control of Burgers equation

Antje Noack · Andrea Walther

Received: 1 December 2003 / Revised: 13 July 2005 / Published online: 21 November 2006
© Springer Science + Business Media, LLC 2006

Abstract Adjoint techniques are a common tool in the numerical treatment of optimal control problems. They are used for efficient evaluations of the gradient of the objective in gradient-based optimization algorithms. Different adjoint techniques for the optimal control of Burgers equation with Neumann boundary control are studied. The methods differ in the point in the numerical algorithm at which the adjoints are incorporated. Discretization methods for the continuous adjoint are discussed and compared with methods applying the application of the discrete adjoint. At the example of the implicit Euler method and the Crank Nicolson method it is shown that a discretization for the adjoint problem that is adjoint to the discretized optimal control problem avoids additional errors in gradient-based optimization algorithms. The approach of discrete adjoints coincides with that of automatic differentiation tools (AD) which provide exact gradient calculations on the discrete level.

Keywords Burgers equation · Optimal control · Automatic differentiation · Adjoint technique · Discrete adjoint · Newton's method

1 Introduction

In the discussion of appropriate numerical methods for optimal control problems governed by partial differential equations, adjoint problem techniques play an essential role. They provide an alternative representation of the gradient of the cost functional which often allows an efficient evaluation of the derivative information required by gradient-based numerical algorithms. In the numerical treatment of optimal control problems a well-balanced combination of linearization and discretization techniques on one hand and of an efficient optimization method on the other hand has to be chosen. The present paper contributes to the discussion

A. Noack (✉)· A. Walther
Institute of Scientific Computing, Technische Universität Dresden, D-01062 Dresden, Germany
e-mail: noack@math.tu-dresden.de

A. Walther
e-mail: awalther@math.tu-dresden.de

of different adjoint problem techniques. The main point of our investigation is the interplay between adjoints and appropriate discretization methods. In particular, we are interested in numerical algorithms which differ in the point at which an adjoint state is used. Adjoint techniques could be used before or after linearization, on a continuous level or on a discrete level. Solving optimal control problems by means of gradient-based optimization algorithms, the convergence depends strongly on the underlying discretization schemes for state and adjoint state. Seperately chosen schemes for state and adjoint which are not adjoint to each other may cause problems. Considering the discrete adjoint, it provides an exact alternative representation of the discretized gradient of the objective. The corresponding scheme for the discrete adjoint can be viewed as a discretization scheme for the continuous adjoint state which is adjoint with respect to the discretized optimal control problem.

There are numerous papers which are concerned with investigations on the optimal control of Burgers equation applying adjoint problem techniques (see e.g. [12, 13, 19–21, 24, 26, 27]). In the present paper, we study a model problem governed by Burgers equation with Neumann boundary control. Burgers equation serves in several fields of research as a one-dimensional model for convection-diffusion phenomena describing e.g. shock waves, traffic flow or certain oceanic phenomena [20, 21]. In reallife problems nonlinearities are a big challenge and require a careful numerical treatment. In nonlinear models the influence of different adjoint techniques is more crucial.

One common way to use adjoints is to construct the continuous adjoint problem as part of the KKT-conditions. The resulting system of equations can then be treated numerically by discretizing each of the equations appropriately [3, 16, 18]. However, as indicated in [20, 21] for a special discretization scheme for Burgers equation, a discretization of the continuous adjoint problem works best if it is adjoint. In the present paper this property is investigated more in general and different adjoint approaches are analyzed.

In [26] the question of appropriate numerical algorithms is discussed in the framework of Hilbert spaces for the unsteady Burgers equation with distributed control, Neumann and homogeneous Dirichlet boundary control. In a quite general setting, discretizations of the state equation and the corresponding discrete Lagrangians are considered. Furthermore, the corresponding discrete adjoint is constructed and an augmented SQP method—a combination of the Lagrange-Newton technique with penalty methods—is applied which proves to work efficiently. The Lagrange—Newton technique will be the first approach which we consider in this paper. We will give a rather compact description referring to [19, 24, 26] for results on convergence.

In the context of automatic differentiation (AD) [12, 13], the reverse mode by which exact discrete gradients can be computed first calculates the discrete adjoint to obtain the gradient in terms of that adjoint. The adjoint codes to compute the gradient are generated automatically from the forward code that evaluates the linearized state equation (see e.g. [10]). The computation of adjoints based on AD coincides with a certain way to implement the discrete adjoint equations. Although divided difference methods would yield the same results, they are quite often prohibitively expensive. Note, that instead of searching for an approximation of the continuous adjoint which refers to its structure, AD considers the discrete adjoints as a mere auxiliary tool to calculate an approximation of the gradient.

The paper is organized as follows. In Section 4, we briefly describe the application of the Lagrange-Newton algorithm to the continuous system. Here the coupling of state and adjoint state gets lost within the first Newton step. But, adjointness is regained in the interpretation from point of view of the SQP method by considering the linearized KKT-system as KKT-system of a sequence of perturbed quadratic optimal control problems. In Section 5,

the state equation is linearized by using Newton's method and the corresponding sequence of Lagrangians is considered. Our main focus lies on appropriate discretization schemes for state and adjoint. This is discussed in Sections 6 and 7. There we consider the question of adjointness of discretization schemes for state and adjoint state. Using an adjoint discretization scheme for the adjoint equation coincides with the application of the discrete adjoint. We recommend this approach since it provides the exact gradient of the discretized objective in every step of the optimization procedure. The adjointness of discretizations is discussed in general and the topic is illustrated at the example of the commonly used implicit Euler method and the Crank-Nicolson method. In Section 8 the discrete adjoint tecnique is applied to the optimal control problem governed by Burgers equation. For investigations on other time integration schemes we refer to [8]. Investigations on conservative schemes for the inviscid Burgers equation can be found in [13]. Some results on numerical tests are stated in Section 9.

2 Formulation of the problem

Consider the optimal control problem

$$\min J(y,u) \quad \text{with} \quad J(y,u)=\frac{1}{2}\int_0^1 (y(x,T;u)-q(x))^2 dx + \frac{1}{2}\int_0^T \big(\alpha_1 u_1^2(t)+\alpha_2 u_2^2(t)\big)dt \tag{1}$$

subject to Burgers equation

$$\begin{aligned} y_t + yy_x - \nu y_{xx} &= 0 && \text{in } \Omega_T = (0,1)\times(0,T)\\ \nu y_x(0,\cdot) &= u_1 && \text{in } (0,T)\\ \nu y_x(1,\cdot) &= u_2 && \text{in } (0,T)\\ y(\cdot,0) &= y^0(\cdot) && \text{in } \Omega = (0,1), \end{aligned} \tag{2}$$

where $\nu > 0$ denotes the viscosity parameter and $T > 0$ the final time. We suppose $y^0 \in L^2(\Omega)$ and $q \in L^2(\Omega)$. For the boundary controls we assume $u = (u_1, u_2) \in (L^2(0,T))^2$.

A lot of work has already been done concerning analytical results for Burgers' equation. In [26] the existence of a unique weak solution for any control $u \in (L_2(0,T))^2$ occurring in boundary conditions of Robin type is proved. That class of problems includes Burgers equation with Neumann boundary conditions as considered here.
In our case the weak solution of (2) is defined in the following sense. Consider the spaces $H = H^1(\Omega)$, $V = L_2(0,1,H)$ and $W(0,T) = \{z \in V \mid z_t \in V^*\}$. Then from [26] we have that there exists a unique weak solution $y \in W(0,T)$ of Burgers Eq. (2) satisfying $y(0) = y^0$ in $L^2(\Omega)$ and

$$\langle y_t(t), \phi\rangle_{H^*,H} + \nu\int_0^1 (y_x(t)\,\phi' + y(t)y_x(t)\phi)dx + u_1(t)\phi(0) - u_2(t)\phi(1) = 0$$
$$\text{for a.e. } t \in (0,T), \quad \text{for all } \phi \in H, \tag{3}$$

where $\langle \cdot, \cdot \rangle_{H^*,H}$ denotes the dual pairing between H and its dual $H^* = (H^1(\Omega))^*$. We define an associated operator $F : W(0,T) \times (L^2(0,T))^2 \to V^*$ by

$$\begin{aligned}\langle F(y,u), \varphi \rangle_{V^*,V} = &\int_0^T \langle y_t(t), \varphi \rangle_{H^*,H}\, dt + \nu \int_0^T \int_0^1 (y_x(t)\,\varphi'(x) + y(t) y_x(t) \varphi(x)) dx dt \\ &+ \int_0^T (u_1(t)\varphi(0) - u_2(t)\varphi(1)) dt \end{aligned} \tag{4}$$

for any $\varphi \in V$. Then the optimal control problem (1)–(2) can be written as

$$\min J(y,u) \quad \text{s.t. } F(y,u) = 0 \text{ with } y \in W(0,T), u = (u_1,u_2) \in (L^2(0,T))^2. \tag{5}$$

Note, that there exists at least one global solution (y^*, u^*) of (5), see [26].

3 Lagrange function and optimality conditions

To solve the optimal control problem (5) we consider the corresponding unrestricted minimization problem based on the Lagrange function $\mathcal{L} : W(0,T) \times (L^2(0,T))^2 \times V \to \mathbb{R}$,

$$\mathcal{L}(y,u,p) := J(y,u) + \langle F(y,u), p \rangle_{V^*,V}, \tag{6}$$

which is given by

$$\min \mathcal{L}(y,u,p) \quad \text{for } (y,u,p) \in W(0,T) \times (L^2(0,T))^2 \times V. \tag{7}$$

Here J and F are twice continuously Fréchet-differentiable and the derivative $\nabla_y F(y,u,v)$ is bijective for any $(y,u) \in W(0,T) \times (L^2(0,T))^2$, see [26]. Now applying a result of [5], these properties ensure the existence of a Lagrange multiplier $p \in V$ such that (y,u,p) solves (7). The following theorem states some further results for the Lagrangian multiplier. In a more general form it is proved in [26].

Theorem 1. *For any solution $(y,u) \in W(0,T) \times (L^2(0,T))^2$ of the optimal control problem (5), there exists a unique function $p \in W(0,T)$ such that (y,u,p) satisfies the first order necessary optimality conditions of (7), i.e. the KKT-system*

$$\begin{aligned} &\nabla_y \mathcal{L}(y^*,u^*,p^*)\eta = 0 \quad \forall \eta \in W(0,T),\ \eta(x,0) = 0,\ \forall x \in (0,1) \\ &\nabla_u \mathcal{L}(y^*,u^*,p^*)v = 0 \quad \forall v \in (L^2(0,T))^2 \\ &\nabla_p \mathcal{L}(y^*,u^*,p^*)\varphi = 0 \quad \forall \varphi \in V. \end{aligned} \tag{8}$$

The Lagrangian multiplier $p \in W(0,T)$ is the unique weak solution of the corresponding adjoint problem

$$\begin{aligned} p_t + yp_x + \nu p_{xx} &= 0 && \text{in } \Omega_T \\ \nu p_x(0,\cdot) + y(0,\cdot)p(0,\cdot) &= 0, && \text{in } (0,T) \\ \nu p_x(1,\cdot) + y(1,\cdot)p(1,\cdot) &= 0, && \text{in } (0,T) \\ p(\cdot,T) &= q(\cdot) - y(\cdot,T) && \text{in } \Omega. \end{aligned} \tag{9}$$

Indeed, the boundary value problem can directly be derived from the first operator equation in the system of KKT-conditions (8) which reads

$$\begin{aligned} \nabla_y \mathcal{L}(y,u,p)\,\eta = &\int_0^1 (y(T)-q)\eta(T)dx + \int_0^T \langle \eta_t, p\rangle_{H^*,H}\,dt \\ &+ \int_0^T\int_0^1 (\eta y_x p + y\eta_x p)dxdt + \nu \int_0^T\int_0^1 \eta_x p_x dxdt = 0 \end{aligned} \tag{10}$$

for any $\eta \in W(0,T)$ that keeps the initial value fixed, i.e. satisfies $\eta(x,0) = 0$, $x \in (0,1)$. Since $p \in W(0,T)$, partial integration can be applied leading to

$$\begin{aligned} \nabla_y \mathcal{L}(y,u,p)\,\eta = &\int_0^1 (y(T)-q)\eta(T)dx + \int_0^1 \eta(T)p(T)dx - \int_0^T \langle \eta, p_t\rangle_{H^*,H} dt \\ &- \int_0^T\int_0^1 \eta y p_x dxdt + \int_0^T (\eta(1)y(1)p(1) - \eta(0)y(0)p(0))dt \\ &+ \nu \int_0^T\int_0^1 \eta_x p_x dxdt = 0. \end{aligned}$$

Now we consider the second equation in (8). It yields

$$\begin{aligned} \nabla_u \mathcal{L}(y,u,p)\,v &= \nabla_u J(y,u)v + \langle \nabla_u F(y,u)v, p\rangle_{V^*,V} \\ &= \alpha_1 \int_0^T u_1(t)\,v_1(t)\,dt + \int_0^T v_1(t)p(0,t)\,dt \\ &\quad + \alpha_2 \int_0^T u_2(t)\,v_2(t)\,dt - \int_0^T v_2(t)p(1,t)\,dt = 0 \end{aligned} \tag{11}$$

for all $v = (v_1, v_2) \in (L^2(0,T))^2$. And thus

$$\alpha_1 u_1(t) + p(0,t) = 0\,, \qquad \alpha_2 u_2(t) - p(1,t) = 0 \quad \text{for } t \in (0,T). \tag{12}$$

They represent the first order necessary optimality condition for (1)–(2). The last equation in (8) coincides with the state equation in variational form (3).

To emphasize the structure we introduce a more compact notation. Consider vectors $w := (y,u) \in X = W(0,T) \times (L^2(0,T))^2$ and denote the derivative w.r.t. w by a prime. Then the first two conditions of System (8) can be formulated as

$$\nabla_w \mathcal{L} = J'(w) + F'(w)^* p = 0, \tag{13}$$

where $F'(w)^* : V \to X^*$ denotes the adjoint operator associated with $F'(w)$.

4 The Lagrange-Newton approach

The numerical approaches considered in this paper have in common that their first step is a linearization step, and that this step is based on a Newton-type iteration to handle the nonlinearity in the problem. The first method on which we focus in this section, is the wellknown Lagrange-Newton method. There the entire KKT-system is linearized by using Newton's algorithm, and the resulting sequence of linear systems is then solved by appropriate numerical techniques. In a slightly different interpretation this method is known as SQP method. The idea of SQP is to define an appropriate sequence of perturbed quadratic optimal control problems corresponding to (1)–(2). These problems are selected such that their KKT-conditions coincide with the linear system which results from the linearization of the original KKT-conditions (cf. [1, 23]). In a variety of articles SQP methods and augmented SQP-methods are applied and analyzed. These papers show that this method works efficiently for a wide class of optimal control problems. For control problems governed by Burgers equation results in this direction can be found e.g. in [17, 24, 26].

We abbreviate $\theta := (y, u, p)$ and use the notation $\nabla^2\mathcal{L}(\theta)$ for the Hessian of the Lagrangian $\mathcal{L}$. Applying Newton's algorithm to the system of KKT-conditions (8) leads to the following sequence of systems of linear equations

$$\nabla^2\mathcal{L}(\theta^{k-1})(\theta^k - \theta^{k-1}) = -\nabla\mathcal{L}(\theta^{k-1}) \quad \text{for } k = 1, 2, \ldots$$

with an appropriate initial value $\theta^0 = (y^0, u^0, p^0) \in W(0,T) \times (L^2(0,T))^2 \times W(0,T)$. From the definitions of $\mathcal{L}$, J and F it follows that this scheme possesses the structure

$$\begin{bmatrix} \nabla_{yy}\mathcal{L}^k & 0 & \nabla_{yp}\mathcal{L}^k \\ 0 & \nabla_{uu}\mathcal{L}^k & \nabla_{up}\mathcal{L}^k \\ \nabla_{py}\mathcal{L}^k & \nabla_{pu}\mathcal{L}^k & 0 \end{bmatrix} \begin{bmatrix} \delta^k y \\ \delta^k u \\ \delta^k p \end{bmatrix} = - \begin{bmatrix} \nabla_y\mathcal{L}^k \\ \nabla_u\mathcal{L}^k \\ \nabla_p\mathcal{L}^k \end{bmatrix}, \quad k = 1, 2, \ldots \tag{14}$$

with the abbreviations $\nabla_{\cdot,\cdot}\mathcal{L}^k = \nabla_{\cdot,\cdot}\mathcal{L}(\theta^{k-1})$, $\delta^k y = y^k - y^{k-1}$, $\delta^k u = u^k - u^{k-1}$, and $\delta^k p = p^k - p^{k-1}$. The first operator equation in (14) represents the linearization of the adjoint Eq. (10). Applying partial integration to that equation and taking some further rearrangements yields

$$\begin{aligned} &\int_0^1 (p^k(T) + y^k(T) - q)\eta(T)\,dx - \int_0^T \langle \eta, p_t^k \rangle_{H^*,H} dt \\ &\quad + \nu \int_0^T\!\!\int_0^1 \eta_x p_x^k dx dt - \int_0^T\!\!\int_0^1 \left(\eta y^{k-1} p_x^k + \eta\, \delta^k y\, p_x^{k-1}\right) dt \\ &\quad + \int_0^T (\eta(1)(y^{k-1}(1)p^k(1) + \delta^k y(1) p^{k-1}(1)) - \eta(0)(y^{k-1}(0)p^k(0) + \delta^k y(0) p^{k-1}(0)))dt = 0 \end{aligned}$$

for all $\eta \in W(0,T)$ with $\eta(x,0) = 0$ for all $x \in (0,1)$ and $k \in \mathbb{N}$. Thus the update p^k can be interpreted as weak solution $p^k \in W(0,T)$ of the BVP

$$\begin{aligned} p_t^k + y^{k-1} p_x^k + \nu p_{xx}^k &= -(y^k - y^{k-1})\, p_x^{k-1} && \text{in } \Omega_T, \\ \nu p_x^k(0,t) + y^{k-1}(0,t) p^k(0,t) &= -(y^k(0,t) - y^{k-1}(0,t))\, p_{k-1}(0,t) && \text{for } t \in (0,T) \\ \nu p_x^k(1,t) + y^{k-1}(1,t) p^k(1,t) &= -(y^k(1,t) - y^{k-1}(1,t))\, p_{k-1}(1,t) && \text{for } t \in (0,T) \\ p^k(x,T) &= q(x) - y^k(x,T) && \text{for } x \in \Omega, \end{aligned} \tag{15}$$

which is a linearization of the continuous adjoint problem (9) at the point p^{k-1}. Observe that the second equation in (14) has the same structure as (12) since the objective J is quadratic and hence its reduced gradient w.r.t. u is linear. It reads

$$\alpha_1 u_1^k(t) + p^k(0,t) = 0\,, \qquad \alpha_2 u_2^k(t) - p^k(1,t) = 0 \quad \text{for } t \in (0,T). \tag{16}$$

The third operator equation in (14) is a linearization of the state equation in the point y^{k-1}. It can be written as classical BVP:

$$\begin{aligned} y_t^k + y^{k-1}\, y_x^k + y_x^{k-1} y^k - \nu y_{xx}^k &= y^{k-1}\, y_x^{k-1} && \text{in } \Omega_T, \\ \nu y_x^k(0,t) &= u_1^k(t) && \text{for } t \in (0,T), \\ \nu y_x^k(1,t) &= u_2^k(t) && \text{for } t \in (0,T), \\ y^k(x,0) &= y^0(x) && \text{for } x \in \Omega. \end{aligned} \tag{17}$$

Under the assumptions made there exists a unique weak solution $y^k \in W(0,T)$ for each $k \in \mathbb{N}$, see e.g. [22].

With the short notation $w^k = (y^k, u^k)$ the Eqs. (15) and (16) can be combined to

$$J'(w^k) + F'(w^{k-1})^* p^k + (F''(w^{k-1})\delta^k w)^* p^{k-1} = 0, \tag{18}$$

which is a linearization of (13).

In [26] the quadratic convergence of the SQP method has been proved for an optimal control problem governed by Burgers equation with Robin boundary conditions. That class of problems includes our case. In [27] an augmented SQP method is studied for Burgers equation.

Remark 1. Considering the linearizations of the adjoint and the state (15) and (17), respectively, one observes that within the first Newton step applied to the KKT system the coupling of state and adjoint gets lost. But, as mentioned above, the character of an optimality system can be maintained by interpreting the Newton iterations from the point of view of the SQP method. There the linearized states and adjoints are considered as KKT conditions of appropriate perturbed quadratic optimal control problems and are re-coupled that way (cf. [1]). To be precise, if we put $w^k = (y^k, u^k)$, $w^{k+1} = (y^{k+1}, u^{k+1})$ and $w = w^{k+1} - w^k$, then in our case the corresponding sequence of quadratic optimal control problems is given by

$$\begin{aligned} \min\ J^k(w^{k+1}) &:= \nabla J(w^k)w + \frac{1}{2}\nabla_{ww}\mathcal{L}(w^k, p^k)(w,w) \\ &= \int_0^1 (y^k(T) - q)(y^{k+1}(T) - y^k(T))\,dx \\ &\quad + \int_0^T \left(\alpha_1 u_1^k\big(u_1^{k+1} - u_1^k\big) + \alpha_2 u_2^k\big(u_2^{k+1} - u_2^k\big)\right)dt \end{aligned}$$

$$+\frac{1}{2}\int_0^1 (y^{k+1}(T)-y^k(T))^2 dx$$
$$+\frac{1}{2}\int_0^T \left(\alpha_1\left(u_1^{k+1}-u_1^k\right)^2+\alpha_2\left(u_2^{k+1}-u_2^k\right)^2\right)dt$$
$$+\int_0^T\int_0^1 p^k(y^{k+1}-y^k)(y^{k+1}-y^k)_x\,dx\,dt \tag{19}$$

such that $y^k(u)$ satisfies (17) for $k \in \mathbb{N}$.

5 The Newton approach applied to the state

Instead of applying Newton's method to the KKT system (8) we apply it now to the nonlinear state Eq. (2) and use the Lagrange technique afterwards. This way a sequence of Lagrange functions is generated associated to the Newton iterates of the linearized optimal control problem. The Lagrange multipliers are the corresponding adjoint states.

Applying Newton's method to the state equation leads to a sequence of control problems of the form

$$\min\ J(y^k,u^k) \quad \text{subject to} \quad L_k(y^k,u^k)=0, \qquad k=1,2,\ldots \tag{20}$$

with an appropriate pair of initial functions $(y^0,u^0)\in W(0,T)\times(L^2(0,T))^2$. Here L_k is the linear operator defined by (17) since the linearized state equation has exactly the same structure as in the first approach. The corresponding sequence of Lagrangians is given by

$$\mathcal{L}_k(y^k,u^k,\varrho^k)=J(y^k,u^k)+\langle L_k(y^k,u^k),\varrho^k\rangle_{V^*,V}.$$

Observe that, by construction and due to linearity, the second and third equation of the KKT systems for the Lagrangians $\mathcal{L}_k(y^k,u^k,\varrho^k)$ have the same structure as the second and third equation of the linearized KKT system (14) of the Lagrange-Newton method, respectively. The first equation in each of the KKT systems represents the adjoint equation of the corresponding linearized optimal control problem. By straightforward calculation it turns out to be equivalent to the BVP

$$\begin{aligned} \varrho_t^k + y^{k-1}\varrho_x^k + \nu\,\varrho_{xx}^k &= 0 && \text{in } \Omega_T,\\ \varrho^k &= 0 && \text{on } \Gamma_T,\\ \varrho^k(x,T) &= q(x)-y^k(x,T) && \forall x\in\Omega \end{aligned} \tag{21}$$

for $k\in\mathbb{N}$. Obviously, in difference to the adjoint of the first method, ϱ^k does not explicitely depend on its predecessor ϱ^{k-1}, due to construction. We have to indicate that although the iterates of the states and controls differ, of course, from those of the Lagrange-Newton method we use simply the same notations y^k and u^k.

Since the optimal control problems in (20) are strictly convex, the existence of a unique optimal point (y_*^k,u_*^k) of (20) is guaranteed for every $k\in\mathbb{N}$.

Collecting the first and second equation of the corresponding KKT-systems in one equation by using the notation $w^k=(y^k,u^k)$, the difference to formula (18) of the Lagrange-Newton

method is best to be seen:

$$J'(w^k) + F'(w^{k-1})^* \varrho^k = 0, \quad k = 1, 2, \dots . \tag{22}$$

In the next section we focus on the numerical treatment of the linearized optimal control problems. In general, to solve the considered optimal control problem numerically appropriate discretizations in space and time have to be chosen. Due to the coupling of state, adjoint state and optimality conditions the discretization methods applied to each of these equations should be related in a certain sense to avoid additional errors. Otherwise it may result in slow convergence of linear system solvers. In the following sections appropriate discretizations shall be discussed.

6 Semi-discretization in space

If discretization methods for the state and the adjoint state equation are chosen appropriately but independently from one another then the adjointness of the two equations is usually broken. Suppose, for example, that adapted grids are applied for the state as well as for the adjoint such that both the equations can be efficiently solved. Then the adjoint coupling gets lost in general due to distinct grids. This may lead to problems if e.g. a gradient-based optimization method is applied and the adjointness is directly used in the algorithm.

This problem is significant likewise for time and space discretizations. We shall discuss it in the next section where time discretization schemes are considered. For discretization in space we simply take piecewise linear C^0-elements on a uniform grid. They are commonly used for numerical tests on Burgers equation. As we will show, they have the property to maintain the adjointness of state and adjoint.

Let a uniform grid $x_i = ih, i = 0, \dots, N$, on $[0, 1]$ be given with $h = 1/N$ denoting the stepsize. Further, let $\varphi_i, i = 0, \dots, N$, denote piecewise linear finite elements on that grid and $V_h = \operatorname{span}\{\varphi_i : i = 0, \dots, N\} \subset H$ the corresponding subspace (for the definition see e.g. [15]). To simplify the notations we introduce abbreviations $\overline{y} := y^{k+1}$, $y := y^k$ and $\overline{u}_i := u_i^{k+1}$, $u_i := u_i^k$, $i = 1, 2$, for the Newton iterates for one of the linearization methods considered above. Then y and $\overline{y}$ are approximated by elements of V_h

$$y(x,t) \sim y_h(x,t) := \sum_{i=0}^{N} y_i(t)\varphi_i(x), \quad \overline{y}(x,t) \sim \overline{y}_h(x,t) := \sum_{i=0}^{N} \overline{y}_i(t)\varphi_i(x) \tag{23}$$

which due to the variational formulation of (17) satisfy

$$\begin{aligned} &\int_0^1 (\overline{y}_h)_t\, \varphi_i \, dx + \int_0^1 (y_h\, \overline{y}_h)_x \varphi_i \, dx + \nu \int_0^1 (\overline{y}_h)_x\, \varphi_i' \, dx \\ &\quad = \int_0^1 y_h\, (y_h)_x\, \varphi_i \, dx + \overline{u}_2(t)\varphi_i(1) - \overline{u}_1(t)\varphi_i(0) \forall\, t \in (0,T), \quad i = 0, \dots, N \end{aligned} \tag{24}$$

$\overline{y}_h(x, 0) = y^0(x) \quad \forall\, x \in [0, 1].$

To keep the notations simple, in the following we identify any $v_h \in V_h$, $v_h(x) \sum_{i=0}^{N} v_i\varphi_i(x)$ with its vector of coefficients $v_h = [v_0, v_1, \dots, v_N]^T$ related to the basis $\{\varphi_i\}$ of V_h, as long as no confusion is to be expected. Then straightforward calculations provide that (24) is

equivalent to the semi-discrete scheme

$$D\overline{y}_h' + \frac{1}{h} A_h \overline{y}_h = \frac{1}{h} f_h(\overline{u}). \tag{25}$$

with matrices

$$C_h := \frac{\nu}{h} \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \dots & 0 & -1 & 1 \end{bmatrix}, \quad D := \frac{1}{6} \begin{bmatrix} 2 & 1 & 0 & \dots & 0 \\ 1 & 4 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & 4 & 1 \\ 0 & \dots & 0 & 1 & 2 \end{bmatrix},$$

$$B_h := \frac{1}{6} \begin{bmatrix} y_1 - 4y_0 & y_0 + 2y_1 & 0 & \dots & 0 \\ -(2y_0 + y_1) & y_2 - y_0 & y_1 + 2y_2 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -(2y_{N-2} + y_{N-1}) & y_N - y_{N-2} & y_{N-1} + 2y_N \\ 0 & \dots & 0 & -(2y_{N-1} + y_N) & 4y_N - y_{N-1} \end{bmatrix}$$

$$f_h := \frac{1}{6} \begin{bmatrix} y_1^2 + y_0 y_1 - 6\overline{u}_1 - 2y_0^2 \\ y_2^2 + y_1(y_2 - y_0) - y_0^2 \\ y_3^2 + y_2(y_3 - y_1) - y_1^2 \\ \vdots \\ y_N^2 + y_{N-1}(y_N - y_{N-2}) - y_{N-2}^2 \\ 2y_N^2 + 6\overline{u}_2 - y_N y_{N-1} - y_{N-1}^2 \end{bmatrix} \quad \text{and} \quad A_h := B_h + C_h. \tag{26}$$

In a similar way we obtain the corresponding semi-discretizations of the adjoint problems (15) and (21) for the Lagrange-Newton method and Newton applied to the state, respectively. Suppose that these adjoints are approximated by functions in V_h:

$$p \sim p_h = \sum_{i=0}^{N} p_i(t)\varphi_i(x), \quad \overline{p} \sim \overline{p}_h = \sum_{i=0}^{N} \overline{p}_i(t)\varphi_i(x) \text{ and } \varrho \sim \varrho_h = \sum_{i=0}^{N} \varrho_i(t)\varphi_i(x),$$

where we used the abbreviations $\overline{p} := \overline{p}^{k+1}$, $p := p^k$ and $\varrho := \varrho^{k+1}$ at Newton step $k+1$. Suppose further that the target function q is approximated by some function $q_h \in V_h$. Then for the Lagrange-Newton method the adjoint Eq. (9) is replaced by the semi-discrete adjoint

scheme

$$\begin{aligned} D\,\overline{p}_h'(t) - \frac{1}{h}A_h^T(t)\,\overline{p}_h(t) &= \frac{1}{h}G_h(t)p_h(t) \\ \overline{p}_h(T) &= q_h - \overline{y}_h(T) \end{aligned} \tag{27}$$

with matrix

$$G_h = \frac{1}{6}\begin{bmatrix} 4\delta_0 y - \delta_1 y & 2\delta_0 y + \delta_1 y & 0 & \dots & 0 \\ -(\delta_0 y + 2\delta_1 y) & \delta_0 y - \delta_2 y & 2\delta_1 y + \delta_2 y & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \dots & \ddots & -(\delta_{N-2} y + 2\delta_{N-1} y) & \delta_{N-2} - \delta_N y & 2\delta_{N-1} y + \delta_N y \\ 0 & \dots & 0 & -(\delta_{N-1} y + 2\delta_N y) & \delta_{N-1} y - 4\delta_N y \end{bmatrix}. \tag{28}$$

Here we used $\delta_i y := \overline{y}_i - y_i$. An equivalent scheme which is forward in time can be obtained by defining $\vartheta := T - t$ and $\hat{\overline{p}}(\vartheta) := \overline{p}(T - \vartheta)$. Then system (27) becomes

$$\begin{aligned} D\,\hat{\overline{p}}_h'(\vartheta) + \frac{1}{h}A_h^T(T-\vartheta)\,\hat{\overline{p}}_h(\vartheta) &= -\frac{1}{h}G_h(T-\vartheta)\hat{p}_h(\vartheta) \\ \hat{\overline{p}}_h(0) &= q_h - \overline{y}_h(T). \end{aligned} \tag{29}$$

For Newton applied to the state the adjoint Eq. (21) is replaced by the semi-discrete scheme

$$\begin{aligned} D\,\varrho_h'(t) - \frac{1}{h}A_h^T(t)\,\varrho_h(t) &= 0 \\ \varrho_h(T) &= q_h - \overline{y}_h(T). \end{aligned} \tag{30}$$

Considered in positive time direction, this system is equivalent to

$$\begin{aligned} D\,\hat{\varrho}_h'(\vartheta) + \frac{1}{h}A_h^T(T-\vartheta)\,\hat{\varrho}_h(\vartheta) &= 0 \\ \hat{\varrho}_h(0) &= q_h - \overline{y}_h(T) \end{aligned} \tag{31}$$

with $\hat{\varrho}_h(\vartheta) := \varrho_h(T - \vartheta)$.

Remark 2. Notice that the two semi-discrete adjoint schemes (27) and (30) are adjoint to the semi-discretization of the linearized optimal control problem corresponding to the Lagrange-Newton method and Newton applied to the state, respectively. Indeed, their solution coincides with the Lagrangian multipliers for the Lagrangian of the corresponding semi-discrete optimal control problems. This is obvious, since for the linear finite element method, all functions y_h, p_h and ϱ_h live in the same space V_h, and the replacement of y, p, ϱ by their approximations

y_h, p_h, ϱ_h in the variational equation maintains the structure. The semi-discretization of the corresponding objectives for the Lagrange-Newton method and Newton applied to the state are constructed in Section 8 below.

To obtain an associated approximation of the reduced gradient of the objective for the two different approaches we indicate that the structure of the formula is the same in both cases. Hence, it suffices to consider one of both. If we take e.g. the Lagrange-Newton method, then from (12) and the approximation of $\overline{p}$ it follows immediately that the associated semi-discretization of the gradient is given by

$$\nabla_u J(\overline{y}, \overline{u}) \approx (\alpha_1 \overline{u}_1 + \overline{p}_0, \alpha_2 \overline{u}_2 - \overline{p}_N). \tag{32}$$

An analogous result is obtained for Newton applied to the state by just replacing $\overline{p}$ with ϱ in the formula.

7 Time integration and discrete adjoint

In this section we discuss appropriate time integration schemes for the adjoint equation. Notice that one can easily select an efficient time integration scheme for the semi-discretization of the state equation as well as of the adjoint equation from a large number of standard numerical methods for differential equations. Euler methods, Crank-Nicolson, Runge-Kutta methods, various multi-step methods are available as part of numerical tools of mathematical software. Such a choice will usually aim for efficiently working schemes for state and adjoint, separately. Hence, in general, it will lead to discrete schemes for the state and the adjoint which are not adjoint anymore, i.e. the corresponding reduced gradient of the discretized objective is not equivalently represented by the formula in terms of the fully discretized adjoint.

Many solvers implemented in mathematical software tools use time step control. Since the solutions of state and adjoint equation often behave rather differently, step size control usually leads to different time grids for the two equations. This is one reason why corresponding discrete solutions of state and adjoint are not adjoint anymore.

In gradient-based optimization algorithms, as e.g. CG-like methods, the solution of the discretized state equation is used to compute the current defect, and the adjoint is applied to calculate the gradient. Hence, applying non-adjoint schemes for state and adjoint may lead to additional errors. Whether these errors are crucial depends on the discretization method chosen and on the character of the optimal control problem.

For the optimal control problem under consideration we discuss and recommend an approach which uses the so-called discrete adjoint. It is constructed with respect to a given discretization for state and objective.

We discuss the pronounced effects first in general and afterwards for two commonly used time integration methods, the implicit Euler method and the Crank-Nicolson method. Both, as it turns out, are not self-adjoint. This effect becomes even more crucial for adapted time steps. Concerning a further discussion of adjointness we refer to [8] where this topic is examined for several other discretization schemes.

Consider an initial value problem of the form

$$F(y, u) = y' - f(y, u) = 0 \ \text{ in } [0, T], \quad y(0) = w \tag{33}$$

with sufficiently smooth right-hand side f, and a corresponding optimal control problem

$$\min J(y,u) \quad \text{subject to} \quad y' - f(y,u) = 0,\ y(0) = w. \tag{34}$$

Notice that in case of the semi-discretization of Burgers Eq. (25) the function on the right-hand side is given by $f(y_h,u) = 1/h\, D^{-1}(F_h - A_h y_h)$.

Applying the Lagrangian technique, the corresponding Lagrangian multiplier, which we denote by λ, is determined as solution of the continuous adjoint equation

$$\nabla_y J + (\nabla_y F)^* \lambda = 0, \qquad \lambda(T) = \lambda_T \tag{35}$$

where $\nabla_y F$ denotes the derivative of F w.r.t. y, and the function λ_T is an appropriate initial condition at final time T.

Suppose now that for time integration some time grid $0 = t^0 < t^1 < \cdots < t^M = T$ with M intervals and step sizes $\tau^j = t^j - t^{j-1}$, $j = 1,\ j = 1,\ldots,M$ is chosen, and a one-step scheme of the form

$$y^{j+1} = y^j + \tau^j \phi(y^j, y^{j+1}, u^j, u^{j+1}) \tag{36}$$

with initial values y^0 and u^0 is selected. Suppose further that the mapping $\phi : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable with respect to all its arguments. Let $y_\tau = [y^1,\ldots,y^M]^T$ and $u_\tau = [u^1,\ldots,u^M]^T$ denote the solution and the control vector, respectively. Then (36) can be written as

$$F^\tau(y^\tau,u^\tau) := \operatorname{diag}((\tau^j)^{-1}) C y^\tau - \Phi(y^\tau,u^\tau) - g = 0$$

with

$$C := \begin{bmatrix} 1 & 0 & \ldots & \ldots & 0 \\ -1 & 1 & \ddots & & \vdots \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \ldots & 0 & -1 & 1 \end{bmatrix}, \quad g = \begin{bmatrix} \frac{y^0}{\tau^1} \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix}$$

and

$$\Phi(y^\tau,u^\tau) = [\phi(y^0,y^1,u^0,u^1),\ldots,\phi(y^{M-1},y^M,u^{M-1},u^M)]^T.$$

Suppose now that $J^\tau(y^\tau,u^\tau)$ is an appropriate approximation of $J(y,u)$ on the given time grid. Then the discrete optimal control problem reads

$$\min J^\tau(y^\tau,u^\tau) \quad \text{subject to} \quad F^\tau(y^\tau,u^\tau) = 0,\ y^0 = w, u^0 = u_0. \tag{37}$$

Considering the Lagrangian

$$\mathcal{L}^\tau(y^\tau,u^\tau,\lambda^\tau) := J^\tau(y^\tau,u^\tau) + \langle F^\tau(y^\tau,u^\tau),\lambda^\tau\rangle_{\mathbb{R}^n}$$

of that problem the KKT-conditions imply the discrete adjoint equation

$$\nabla_y J^\tau + \big(\mathrm{diag}((\tau^j)^{-1})C^T - (\nabla_y \Phi)^T\big)\lambda^\tau = 0 \tag{38}$$

defining the corresponding Lagrangian multiplier $\lambda^\tau = [\lambda^0, \ldots, \lambda^{M-1}]$. Let us assume that $f(y, u)$ is such that for every control u^τ a solution $y^\tau(u^\tau)$ exists. Then we may introduce the function $\mathcal{J}^\tau(u^\tau) := J^\tau(y^\tau(u^\tau), u^\tau)$ and from the first order necessary optimality condition, which is part of the KKT system, we receive an alternative representation of the discrete reduced gradient of the objective given by

$$\mathcal{J}'(u)\eta = \nabla_u J^\tau \eta - \langle \nabla_u \Phi\, \eta, \lambda^\tau \rangle_{\mathbb{R}^n} = 0.$$

Remark 3. Due to the structure of $\nabla_y \Phi$, the adjoint equation can be interpreted as discretization scheme

$$\begin{aligned}
&\frac{1}{\tau^j}(\lambda^{j-1} - \lambda^j) - ((\nabla_x \phi)^T(y^{j-1}, y^j, u^{j-1}, u^j)\lambda^{j-1} + (\nabla_y \phi)^T(y^j, y^{j+1}, u^j, u^{j+1})\lambda^j) \\
&\quad = -[\nabla_y J^\tau]_j \quad j = M-1, \ldots, 1 \\
&\frac{1}{\tau^j}(\lambda^{M-1} - \lambda^M) - \big((\nabla_u \phi)^T(y^{M-1}, y^M, u^{M-1}, u^M)\lambda^{M-1}\big) = 0 \quad \lambda^M = -[\nabla_y J^\tau]_M.
\end{aligned} \tag{39}$$

for the continuous adjoint problem (35). Here $[\nabla_y J^\tau]_k$ denotes the kth component of $\nabla_y J^\tau$. Observe that, by construction, the time grid in (39) coincides with that of the state equation. Hence, if time step control is applied, the grid is adapted to the behaviour of the states and not to that of the adjoint. So, in general, this scheme does not provide an efficient approximation of the continuous adjoint. But it is actually not intended to be one. The discrete adjoint should not primarily be interpreted as approximation of the continuous adjoint but as an auxiliary tool to compute the reduced gradient of the discretized objective in an efficiently way.

Often in applications the same discretization scheme is applied for state and adjoint state with possibly adapted time grids. Therefore, the question arises whether commonly used methods for time as well as for spatial discretizations are self-adjoint. Due to the definition given in [13] we call a numerical discretization method *self-adjoint* if its application to the corresponding continuous adjoint equation results in a scheme which coincides with the scheme for the discrete adjoint.

We want to illustrate the problem of self-adjointness by considering two well-known time integration methods, the implicit Euler method and the Crank-Nicolson method. We show that both are not self-adjoint.

The implicit Euler method is often applied to evolution equations, in particular to stiff systems. For Burgers equation it is commonly used. With $\phi(y^j, y^{j+1}, u^j, u^{j+1}) := f(y^{j+1}, u^{j+1})$ it is given by

$$\mathrm{diag}\big((\tau^j)^{-1}\big)Cy^\tau - f^\tau(y^\tau, u^\tau) - g = 0$$

with

$$f^\tau := [f^1, \dots, f^M]^T \quad \text{and} \quad f^j(y^\tau, u^\tau) := f(y^j, u^j)$$

and g as above. Due to (39) the discrete adjoint that corresponds to implicit Euler is the solution of the system

$$\frac{1}{\tau_j}(\lambda^{j-1} - \lambda^j) - (\nabla_y f)^T(y^j, u^j)\lambda^{j-1} = -[\nabla_y J^\tau]_j, \quad j = M, \dots, 1$$
$$\lambda^M = -[\nabla_y J^\tau]_M. \tag{40}$$

Thus, the implicit Euler method is adjoint to implicit Euler with frozen coefficients on the time grid for the state equation. In this case the alternative representation of the discrete reduced gradient is given by

$$\mathcal{J}'(u)\eta = \nabla_u J^\tau \eta - \langle \nabla_y f^\tau \eta, \lambda^\tau \rangle = 0,$$

It is obvious that if we take an approximation of the continuous adjoint by means of implicit Euler instead of the discrete adjoint, the difference in the discretized reduced gradient is not of relevance for fine grids as long as a uniform time grid is chosen. It becomes more crucial for adapted time grids.

As a second example we want to discuss the Crank-Nicolson method which is determined by the iteration scheme

$$y^{j+1} = y^j + \frac{\tau^j}{2}(f(y^{j+1}, u^{j+1}) + f(y^j, u^j)),$$

i.e., here we have $\phi(y^j, y^{j+1}, u^j, u^{j+1}) = 1/2(f(y^{j+1}, u^{j+1}) + f(y^j, u^j))$. Then for the corresponding discrete adjoint we obtain the scheme

$$\frac{1}{\tau_j}(\lambda^{j-1} - \lambda^j) - \frac{1}{2}(\nabla_y f)^T(y^j, u^j)(\lambda^{j-1} + \lambda^j) = -[\nabla_y J^\tau]_j, \quad j = M, \dots, 1$$
$$\frac{1}{\tau_M}(\lambda^{M-1} - \lambda^M) - \frac{1}{2}(\nabla_y f)^T(y^M, u^M)\lambda^{M-1} = 0 \quad \lambda^M = -[\nabla_y J^\tau]_M. \tag{41}$$

In difference, applying the Crank-Nicolson scheme to the continuous adjoint leads to the scheme

$$\frac{1}{\tau_j}(\lambda^{j-1} - \lambda^j) - \frac{1}{2}((\nabla_y f)^T(y^{j-1})\lambda^{j-1} + (\nabla_y f)^T(y^j)\lambda^j) = -[\nabla_y J^\tau]_j, \quad j = M, \dots, 1$$
$$\lambda^M = -[\nabla_y J^\tau]_M.$$

Via the discrete adjoint an exact gradient of the discretized objective is computed. This adjoint technique coincides with the corresponding concept in automatic differentiation (AD), also called algorithmic differentiation. AD offers an opportunity to provide derivative information for a given code segment by applying the chain rule systematically to statements of computer programms. For that purpose, the code is decomposed into a typically very

long sequence of simple evaluations, e.g. additions, multiplications, and calls to elementary function such as $\sin(x_i)$ or $\exp(x_i)$. The derivatives with respect to the arguments of these operations can be easily calculated. Furthermore, the chain rule yields the derivatives of the whole sequence with respect to the input variables. Depending on the starting point of this methodology—either at the beginning or at the end of the respective chain of computational steps—one distinguishes between the forward mode and the reverse mode of AD.

Since the reverse mode computes the exact gradient information of the given computer program, it can be viewed as a discrete analog to the application of the continuous adjoint equation to the reduced gradient of the continuous optimal control problem. These parallels have already been hinted at in [9]. The time-complexity result for the reverse mode is based on the operation count of the underlying function. The gradient of a scalar-valued component function is obtained using the reverse mode in its basic form at no more than five times operation count of the underlying function [10].

The memory requirement of the basic reverse mode is proportional to the time needed to evaluate the function F itself. However, there are several strategies based on checkpointing to reduce this memory requirement and a remarkable decrease can be obtained effortless. Moreover, for the time-integration problem considered in this paper there exist optimal checkpointing schedules that allow a logarithmic increase in the memory requirement [12]. With respect to checkpointing, in [12, 13] Burgers equation is investigated and the quality of the approximation of the gradient is discussed. There in difference to our approach discretizations of the original BVP as e.g. Lax-Friedrichs flow and Godunow flow are applied without a linearization of the state equation. In the next section we construct the discrete adjoint corresponding to implicit Euler for the semi-discretized optimal control problem governed by Burgers equation.

8 Discrete adjoints for Burgers equation

In this section we return to the optimal control problem (1)–(2). For the two linearization methods, the Lagrange-Newton method and the Newton method applied to the state discrete adjoints shall be constructed, and an alternative representation of the reduced gradient of the discretized objective shall be derived. The methods require a slightly different treatment since for the Lagrange-Newton method the adjoint of the preceding Newton step occurs in the objective whereas for the Newton method applied to the state it does not.

Recall that applying the Lagrange-Newton method leads to the sequence of quadratic optimal control problems given in (19). Associated to the spatial discretization of states and adjoint states we use linear finite elements to derive an appropriate semi-discretization of the objective. That means to replace $\overline{y} := y^{j+1}$ and $y := y^j$ by its finite element approximations $\overline{y}_h$, $y_h \in V_h$, respectively, in (19) and yields

$$\begin{aligned} J^k(\overline{y}_h, \overline{u}) \approx J_h^k(\overline{y}_h, \overline{u}) &:= h(\overline{y}_h(T) - q_h)^T D(\overline{y}_h(T) - y_h(T)) \\ &+ \int_0^T (\alpha_1 u_1(\overline{u}_1 - u_1) + \alpha_2 u_2(\overline{u}_2 - u_2))dt + \frac{h}{2}(\overline{y}_h(T) - y_h(T))^T D(\overline{y}_h(T) - y_h(T)) \\ &+ \frac{1}{2}\int_0^T (\alpha_1(\overline{u}_1 - u_1)^2 + \alpha_2(\overline{u}_2 - u_2)^2)dt + \int_0^T (\overline{y} - y)^T P_h(\overline{y} - y)\, dt \end{aligned} \tag{42}$$

with matrix D as defined in (26) and a matrix $P_h = P_h(p_h)$ that depends on the preceding semi-discretized adjoint $p_h = [p_0, \dots, p_N]^T$ and is defined by

$$P_h = \frac{1}{6}\begin{bmatrix} -(2p_0+p_1) & 2p_0+p_1 & 0 & \dots & 0 \\ -(p_0+2p_1) & p_0-p_2 & 2p_1+p_2 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \dots & \ddots & -(p_{N-2}+2p_{N-1}) & p_{N-2}-p_N & 2p_{N-1}+p_N \\ 0 & \dots & 0 & -(p_{N-1}+2p_N & p_{N-1}+2p_N \end{bmatrix}. \quad (43)$$

For time integration let be given a time grid $0 = t_0 < t_1 < \cdots < t_M = T$ with step sizes $\tau^j = t^j - t^{j-1}$, $j = 1, \dots, M$, then applying implicit Euler to the semi-discretized state Eq. (25) for a discretized control $\overline{u}^\tau := [\overline{u}_1^\tau, \overline{u}_2^\tau]$, $u_i^\tau = [u_i^1, \dots, u_i^M]$ for $i = 1, 2.$ leads to discrete schemes

$$D\,\frac{\overline{y}^j - \overline{y}^{j-1}}{\tau^j} + \frac{1}{h}A_h^j\,\overline{y}^j = \frac{1}{h}f_h^j(\overline{u}^j), \quad j = 1, \dots, M \quad (44)$$

for the corresponding vector $y_\tau = [y^1, \dots, y^M]^T$, where $A_h^j := A_h(y^j)$ and $f_h^j = f_h(y^j, \overline{u}^j)$. To obtain an appropriate time discretization of the semi-discretized objective (42) the integrals in the formula are in a natural way approximated by a sum, discrete controls $\overline{u}^\tau := [\overline{u}_1^\tau, \overline{u}_2^\tau]$ and the time discretizations y_τ, z_τ of y_h and z_h are used.

Moreover we have to incorporate an appropriate discretization of p_h since its values determine the entries in the matrix P_h. Observe that this approximation is needed to obtain a discretization of the objective and thus is different from applying the adjoint for an alternative representation of a discrete gradient. Therefore we could select any efficient approximation independent of our choice to take the discrete adjoint for representing the gradient. Selecting an adapted grid in that approximation only requires that the result has to be transformed to the given time grid for the state. Here, however, we choose the discrete adjoint also for approximating p_h in (42). That way computed values are re-used in the algorithm. Only in the first step of the Newton iterations an initial adjoint has to be selected.

The time discretization of the semi-discretized objective reads

$$\begin{aligned} J_h^k(\overline{y}_h, \overline{u}) \approx J_{h,\tau}^k(\overline{y}_\tau, \overline{u}_\tau) &:= h(\overline{y}^M - q_h)^T D(\overline{y}^M - y^M) + \frac{h}{2}(\overline{y}^M - y^M)^T D(\overline{y}^M - y^M) \\ &+ \sum_{j=1}^M \tau^j \left(\alpha_1 u_1^j\left(\overline{u}_1^j - u_1^j\right) + \alpha_2 u_2^j\left(\overline{u}_2^j - u_2^j\right)\right) \\ &+ \frac{1}{2}\sum_{j=1}^M \tau^j \left(\alpha_1\left(\overline{u}_1^j - u_1^j\right)^2 + \alpha_2\left(\overline{u}_2^j - u_2^j\right)^2\right) + \sum_{j=1}^M \tau^j (\overline{y}^j - y^j)^T P^j (\overline{y}^j - y^j), \end{aligned} \quad (45)$$

where P^j equals P_h when the entries in terms of p are replaced by the adequate entries in terms of the discrete adjoint of the previous Newton step $\lambda = [\lambda^0, \dots, \lambda^{M-1}] \in \mathbb{R}^{M(N+1)}$, which is specified below. For $k = 0$ an appropriate initial approximate adjoint has to be chosen since no discrete adjoint is available sofar.

If we introduce the notation $\mathcal{J}^k_{h,\tau}(u^\tau) := J^k_{h,\tau}(y^\tau(u^\tau), u^\tau)$, where $y^\tau(u^\tau)$ is the response of (44) to the control u^τ, then its gradient in direction of $\eta = [\eta_1, \eta_2] \in I\!R^M \times I\!R^M$ is of the form

$$\begin{aligned}\left(\mathcal{J}^k_{h,\tau}\right)'(\overline{u}^\tau)\eta = h(\overline{y}^M - q_h)^T D z^M + \sum_{j=1}^{M} \tau^j (\overline{y}^j - y^j)^T (P^j + (P^j)^T) z^j \\ + \sum_{j=1}^{M} \tau^j \left(\alpha_1 \overline{u}_1^j \eta_1^j + \alpha_2 \overline{u}_2^j \eta_2^j\right),\end{aligned} \tag{46}$$

where $z^\tau = z^\tau(\eta) = [z^1, \ldots, z^M]^T \in I\!R^{M(N+1)}$ is the solution of the equation for the sensitivities associated to y^τ:

$$D\,\frac{z^j - z^{j-1}}{\tau^j} + \frac{1}{h} A_h^j z^j = \frac{1}{h}\left(\eta_2^j\, e^{N+1} - \eta_1^j\, e^1\right), \quad j = 1, \ldots, M, z^0 = 0 \tag{47}$$

with e^1 and e^{N+1} denoting the first and $(N+1)$st unit vectors in $I\!R^{N+1}$, respectively. The discrete adjoint equation corresponding to the discretized optimal control problem results from (40) for the special situation of the linearized and semi-discretized Burgers equation. We obtain

$$\begin{aligned}D\frac{1}{\tau^j}(\lambda^{j-1} - \lambda^j) + \frac{1}{h}\left(A_h^j\right)^T \lambda^{j-1} = -(P^j + (P^j)^T)(\overline{y}^j - y^j), \quad j = M, \ldots, 1 \\ \lambda^M = -h(\overline{y}^M - q_h).\end{aligned} \tag{48}$$

To find an alternative expression of the discrete reduced gradient of the objective by means of the discrete adjoint we introduce the matrix $Q \in \mathcal{L}(I\!R^{M(N+1)})$ given by

$$Q = \begin{pmatrix} D + \beta_1 A_h^1 & 0 & \cdots & \cdots & 0 \\ -D & D + \beta_2 A_h^2 & 0 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -D & D + \beta_M A_h^M \end{pmatrix}$$

with $\beta_i = \tau^i / h$ for $i = 1, \ldots, M$. Then (47) is equivalent to the matrix equation

$$Q\, z = \mathrm{diag}(\beta_j)\, b, \qquad b = [b_1, \ldots, b_M],\ b_j := \eta_2^j e^{N+1} - \eta_1^j e^1,\ j = 1, \ldots, M.$$

Multiplying this equation by the discrete adjoint vector $\lambda := [\lambda^0, \ldots, \lambda^{M-1}]^T \in \prod_{l=1}^{M} I\!R^{N+1}$ from the right and using (48) leads to

$$\begin{aligned}h(\overline{y}^M - q_h)^T D z_h^M + \sum_{j=1}^{M} \tau^j (\overline{y}^j - y^j)^T (P^j + (P^j)^T) z^j &= -(Q^T\lambda)^T z = -\lambda^T Q z \\ &= -\mathrm{diag}(\beta_k)\lambda^T b.\end{aligned}$$

Therefore the discrete gradient becomes

$$\left(\mathcal{J}_{h,\tau}^{k}\right)'(\overline{u}_\tau)\eta = \sum_{j=1}^{M} \tau^j \left(\left[\frac{1}{h}\lambda_0^j + \alpha_1 \overline{u}_1^j \right] \eta_1^j + \left[-\frac{1}{h}\lambda_N^j + \alpha_2 \overline{u}_2^j \right] \eta_2^j \right) \tag{49}$$

for any $\eta = (\eta_1, \eta_2) \in I\!R^M \times I\!R^M$.

Now we consider the other approach. For Newton's method applied to the state the semi-discretization of the original objective has to be considered. Using linear finite elements it reads

$$J(\overline{y}, \overline{u}) \approx J_h(\overline{y}_h, \overline{u}) := \frac{h}{2}(\overline{y}_h(T) - q_h))^T D(\overline{y}_h(T) - q_h) + \frac{1}{2}\int_0^T \left(\alpha_1 \overline{u}_1^2(t) + \alpha_2 \overline{u}_2^2(t)\right) dt, \tag{50}$$

and applying an appropriate time discretization analogously as for the Lagrange-Newton method it becomes

$$J_h(\overline{y}_h, \overline{u}) \approx J_{h,\tau}(\overline{y}^\tau, \overline{u}^\tau) := h\left(\overline{y}^M - q_h\right)^T D + \frac{1}{2}\sum_{j=1}^{M} \tau^j \left(\alpha_1 \left(\overline{u}_1^j\right)^2(t) + \alpha_2 \left(\overline{u}_2^j\right)^2\right). \tag{51}$$

With the notation $\mathcal{J}_{h,\tau}(u^\tau) := J_{h,\tau}(y^\tau(u^\tau), u^\tau)$ for the reduced gradient we then have

$$(\mathcal{J}_{h,\tau})'(\overline{u}^\tau)\eta = h\left(\overline{y}^M - q_h\right)^T D z^M + \sum_{j=1}^{M} \tau^j \left(\alpha_1 \overline{u}_1^j \eta_1^j + \alpha_2 \overline{u}_2^j \eta_2^j\right) \tag{52}$$

for any direction $\eta = (\eta_1, \eta_2) \in I\!R^M \times I\!R^M$. The discrete adjoint equation results from (40). Denoting the discrete adjoint by ψ, (40) yields

$$\begin{aligned} D\frac{1}{\tau^j}(\psi^{j-1} - \psi^j) + \frac{1}{h}\left(A_h^j\right)^T \psi^{j-1} = 0, \quad j = M, \ldots, 1 \\ \psi^M = -h(\overline{y}^M - q_h). \end{aligned} \tag{53}$$

The representation of the discrete gradient $(\mathcal{J}_{h,\tau})'(\overline{u}_\tau)$ in terms of the discrete adjoint can be derived similarly as for the Lagrange-Newton method and has exactly the same structure as (49). It only differs in that λ is replaced by ψ:

$$(\mathcal{J}_{h,\tau})'(\overline{u}_\tau)\eta = \sum_{j=1}^{M} \tau^j \left(\left[\frac{1}{h}\psi_0^j + \alpha_1 \overline{u}_1^j \right] \eta_1^j + \left[-\frac{1}{h}\psi_N^j + \alpha_2 \overline{u}_2^j \right] \eta_2^j \right). \tag{54}$$

9 Numerical tests

For an optimal control problem governed by Burgers equation we derived appropriate representations of the reduced gradient of the discretized objective for two different linearization methods, the Lagrange-Newton method and the Newton method applied to the state. The formulas in both approaches allow an exact and efficient calculation of the reduced gradient in terms of the discrete adjoint.

An implementation in a computer program can be either done by hand-coding or automatically be generated by applying an AD-tool. There are well-tested AD-tools available for all important programming languages, e.g. ADIFOR (Argonne National Laboratory, [4]) and TAF (FastOpt, [7]) for Fortran-codes, ADOL-C (Technische Universität Dresden, [11]) and FADBAD (Technical University of Denmark, [2]) for C/C++-programms as well as ADMAT (Cornell University, [25]) for MATLAB. Using one of these tools offers the possibility to avoid the time-consuming and error-prone task of producing a program to evaluate exact discrete derivatives by hand.

In the following the discrete adjoint technique shall be tested by means of numerical examples. We illustrate it for Newton applied to the state. The Lagrange-Newton method provides similar results but needs more computing time and more storage since the formulas for the gradient as well as the discrete adjoint include additional terms where the adjoint of the previous Newton iteration is incorporated.

Further numerical results for Burgers equation, using other linearization methods like reduced SQP methods and alternative methods for solving the discrete linear systems, can be found in e.g. [3, 18, 26].

As semi-discretization in space we take linear finite elements on a uniform grid with step size $h = 1/N$. For time integration we choose implicit Euler on a uniform grid with step size $\tau = 1/M$. To solve the linear systems that occur in each Newton step we select a CG algorithm. As starting value y^0 for the Newton iterations we choose a function which is constant with respect to time and coincides with the target q for any value of $x \in [0, 1]$. The results of the numerical examples show that only a small number of Newton steps is required to obtain a good approximation of the optimal solution. This is due to the quadratic convergence of Newton's algorithm.

All the examples are implemented in MATLAB 7.0.

Example 1. Consider as target $q(x)$ the solution $y(x, t; u)$ of Burgers equation at final time $t = T$ for boundary controls

$$u_1(t) = \sin(2\pi t), \quad u_2(t) = 0 \quad \text{for all } t \in [0, 1]$$

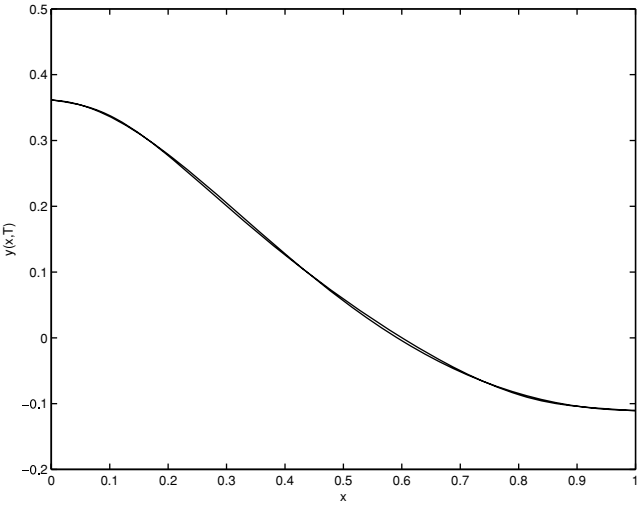


Fig. 1 Optimal solution versus target

and inital value $y_0(x) = 0$ for all $x \in [0, 1]$. It is computed by using an intrinsic MATLAB function with a uniform grid 10-times finer then the grid used for optimization. As parameters for the discretization we select $M = 500$, $N = 500$, $\nu = 0.5$ and $\alpha = 1e - 3$. The optimal solution at final time, the corresponding optimal control and a surface plot of the optimal solution are given in Figs. 1–3.

It needs only one Newton step to obtain an approximation of the optimal solution with accuracy of the order of the discretization error. The error after one Newton step is about 0.00263.

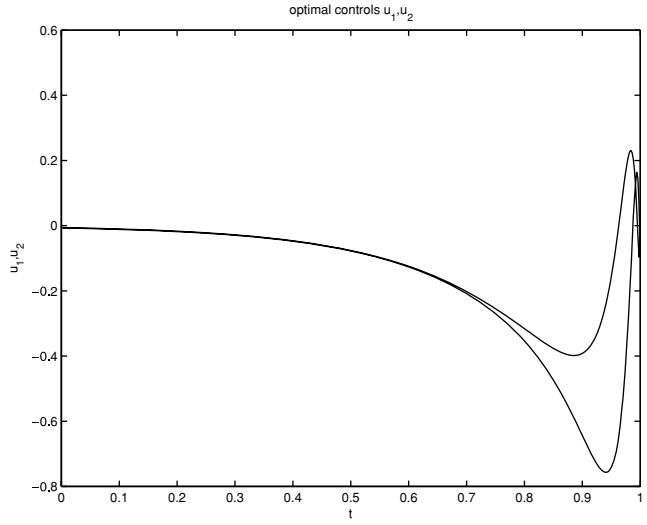
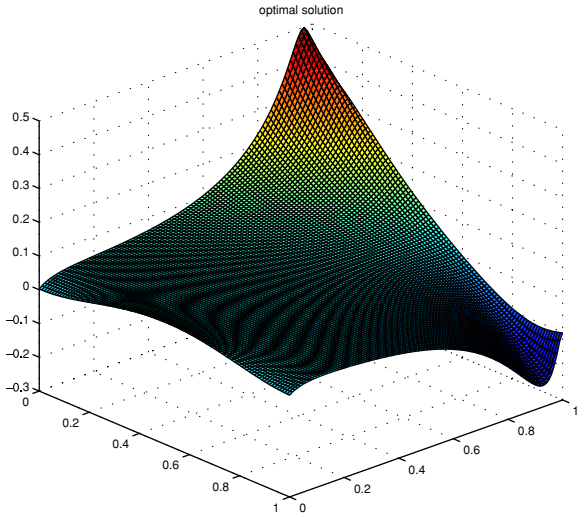


Fig. 2 Optimal controls u_1 and u_2

Fig. 3 Optimal solution



Example 2. As target at final time $t = T$ we select the function

$$z(x) = \begin{cases} 2x & \text{for } x \leq 0.5 \\ 2(1-x) & \text{for } x > 0.5. \end{cases}$$

We choose parameters $M = 500$, $N = 500$, $\nu = 0.5$ and $\alpha = 1e-3$ and take $y_0(x) = 0$ for all $x \in [0, 1]$. The results are plotted in Figs. 4–6.

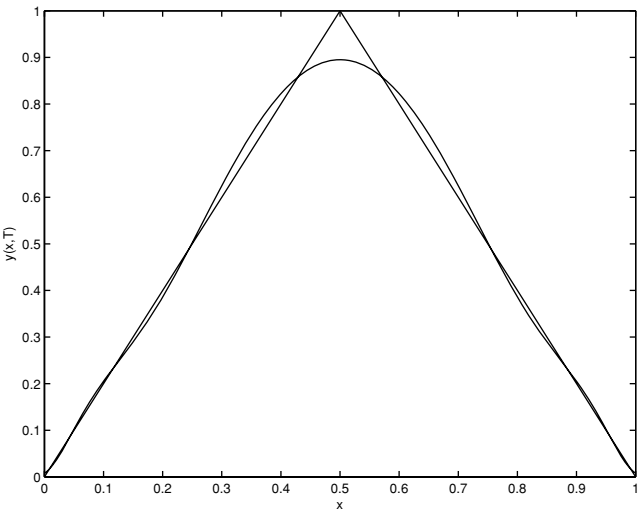


Fig. 4 Optimal solution versus target

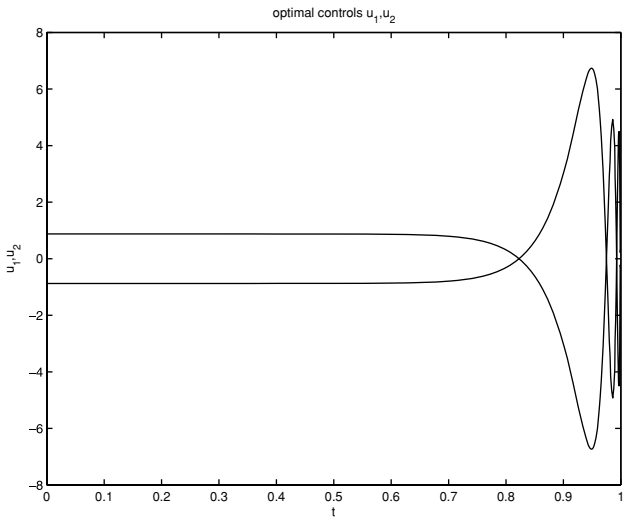
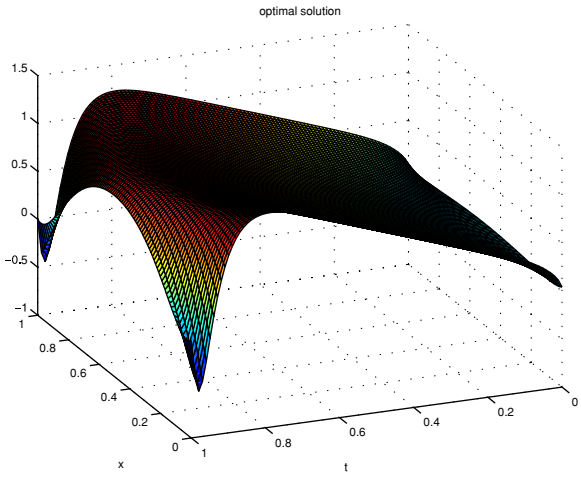


Fig. 5 Optimal controls u_1 and u_2

Due to the quadratic convergence of Newton, here it needs only two steps of the Newton algorithm to obtain an approximation of the optimal solution of order of the discretization error. The error after two Newton steps is about 0.0434.

Example 3. We now choose a smaller value $\nu = 0.1$ for the viscosity and parameters $M = 1000$, $N = 500$, and $\alpha = 1e-4$. Further we select a constant target $z(x) = 0.035$ and $y_0 = 0$ for all $x \in [0, 1]$ (the example is taken from [27]). The optimal control and the optimal solution are shown in Figs. 7 and 8, respectively. After two steps of the Newton algorithm the error between approximate optimal solution at final time and the target function is about $3.36e-6$.

Fig. 6 Optimal solution



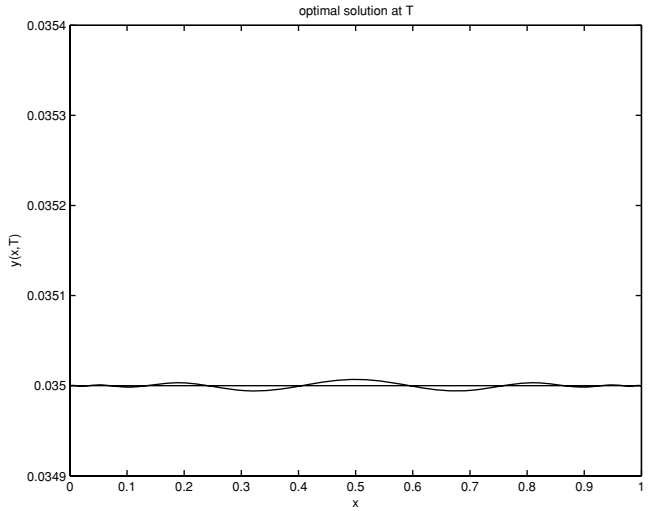


Fig. 7 Optimal controls u_1 and u_2

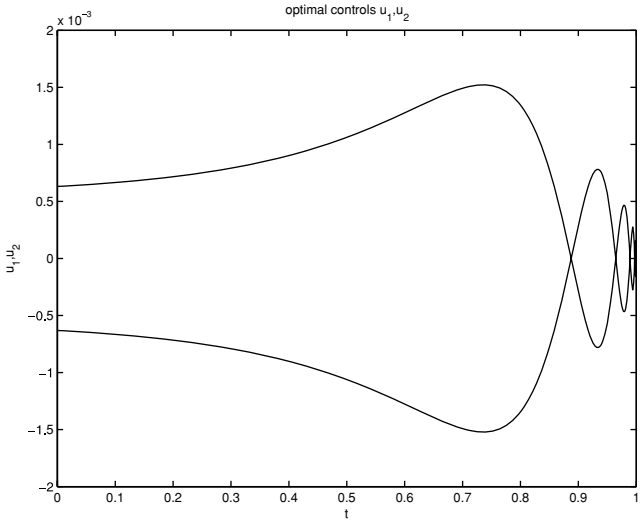


Fig. 8 Optimal solution at $t = T$ and target q

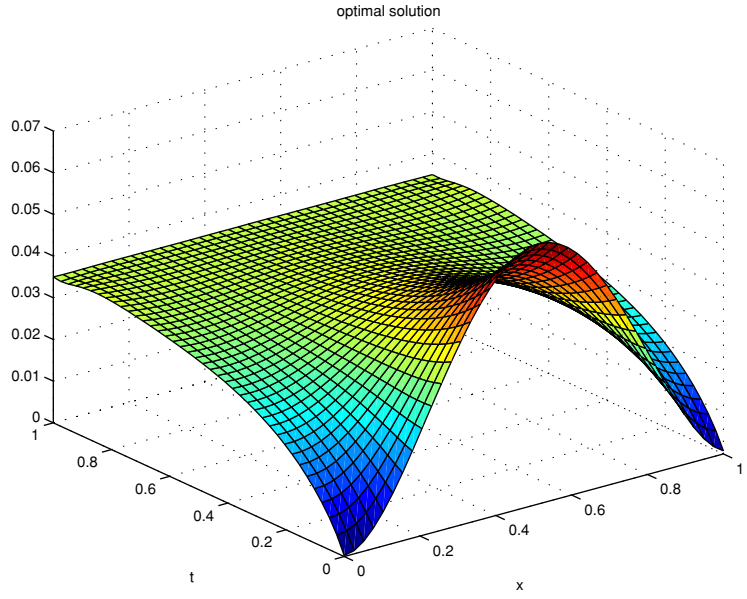


Fig. 9 Optimal solution

We are aware of the fact that the considered problems are ill-posed. Therefore an appropriate preconditioning would lead to better convergence of the CG algorithm. But for an illustration of the discussed effects we simply used the basic numerical algorithm.

Acknowledgments We thank Ch. Grossmann and A. Griewank for fruitful discussions. We kindly acknowledge that this work was partially sponsored by the DFG grants GR 1777/2-2.

References

1. W. Alt, "The Lagrange-Newton method for infinite-dimensional optimization problems," Numer. Func. Anal. and Optim., vol. 11, pp. 201–224, 1990.
2. C. Bendtsen and O. Stauning, "FADBAD, a flexible C++ package for automatic differentiation," Technical University of Denmark, IMM, Technical Report IMM-REP-1996-17 (1996).
3. G. Biros and O. Ghattas, "Parallel Lagrange-Newton-Krylov-Schur methods for PDE-constrained optimization. Part I: the Krylov-Schur solver," SIAM Journal on Scientific Computing, to appear.
4. C. Bischof, A. Carle, P. Khademi, A. Maurer, and P. Hovland, ADIFOR 2.0 User's Guide, ANL. ANL/MCS-TM-192, 1994.
5. P.G. Ciarlet, Introduction to Numerical Linear Algebra and Optimization, Cambridge Univers. Press, 1989.
6. R. Dautray and J.-L. Lions, Mathematical Analysis and Numerical Methods for Science and Technology. Volume 5: Evolution Problems I, Springer Verlag, 1992.
7. R. Giering and T. Kaminski, "Recies for adjoint code construction," ACM Trans. Math. Software, vol. 24, pp. 437–474, 1998.
8. R. Griesse and A. Walther, "Evaluating gradients in optimal control—continuous adjoints versus automatic differentiation," J. Optim. Theory and Appl.(JOTA), vol. 122, no. 1, pp. 63–86, 2004.
9. A. Griewank, "On automatic differentiation," in Mathematical Programming: Recent Developments and Applications, (Eds.), M. Iri and K. Tanabe, Kluwer Academic Publishers, 1989, pp 83–107.
10. A. Griewank, "Evaluating derivatives: Principles and techniques of algorithmic differentiation," Frontiers in appl. math vol. 19, 2000.
11. A. Griewank, D. Juedes, and J. Utke, "ADOL–C, a package for the automatic differentiation of algorithms written in C/C++," ACM Trans. Math. Soft., vol. 22, pp. 131–167, 1996.
12. A. Griewank and A. Walther, "Revolve: An implementation of checkpointing for the reverse or adjoint mode of computational differentiation," TOMS, vol. 26, no. 1, 2000.
13. A. Griewank and A. Walther, "Applying the checkpointing routine treeverse to discretizations of Burgers' equation," Lect. Notes Comput. Sci.and Engin. 8: in High Performance Scientific and Engineering Computing, H.-J. Bungartz, F. Durst, and C. Zenger (Eds.) Springer Berlin Heidelberg, 1999.
14. Ch. Grossmann and A. Noack, "Linearizations and adjoints of operator equations – constructions and selected applications," Preprint MATH-NM-08-01, TU Dresden, 2001.
15. Ch. Grossmann and H.-G. Roos, "Numerik partieller Differentialgleichungen," Teubner, 1992.
16. M. Hinze, "Optimal and instantaneous control of the instationary Navier-Stokes equations," Habilitationsschrift, Fachbereich Mathematik, Technische Universität Berlin, 1999.
17. M. Hinze and S. Volkwein, "Analysis of instantaneous control for the Burgers equation," Nonlinear Anal. TMA 50A, no. 1, pp. 1–26, 2002.
18. D. Keyes, P. Hovland, L. McInnes, and W. Samyono, "Using automatic differentiation for second-order matrix-free methods in pde-constrained optimization," in Automatic Differentiation of Algorithms: From Simulation to Optimization, G. Corliss, C. Faure, A. Griewank, L. Hascoet, and U. Naumann (Eds.), Springer, New York, 2001.
19. K. Kunisch and S. Volkwein, "Control of Burgers' equation by a reduced order approach using proper orthogonal decomposition," J. Optim. Theory and Appl, vol. 102, pp. 345–371, 1999.
20. J.-M. Lellouche, J.-L. Devenon, and I. Dekeyser, "Boundary control of Burgers' equation—a numerical approach," Comput Math. Applic., vol. 28, no. 5, pp. 33–44, 1994.
21. Y. Leredde, J.-M. Lellouche, J.-L. Devenon, and I. Dekeyser, "On initial, boundary conditions and viscosity coefficient control for Burgers' equation," Int. J. Numer. Meth. Fluids, vol. 28, pp. 113–128, 1998.
22. R. Temam, Navier-Stokes Equations," Studies in Mathematics and its Applications. North-Holland, 1979.
23. F. Tröltzsch, "On the Lagrange-Newton-SQP method for the optimal control of semilinear parabolic equations," SIAM J. Control Optimization, vol. 38, no. 1, pp. 294–312, 1999.
24. F. Tröltzsch, and S. Volkwein, "The SQP method for control constrained optimal control of the Burgers equation," ESAIM: Control, Optimisation and Calculus of Variations, vol. 6, pp. 649–674, 2001.
25. A. Verma, "ADMAT: An automatic differentiation and MATLAB interface toolbox," in Object-Oriented Methods for Interoperable Scientific and Engineering Computing, M. Henderson, C. Anderson, and S. Lyons (Eds.), SIAM, 1999, pp. 174–183.
26. S. Volkwein, "Second order conditions for boundary control problems of the Burgers equation," Control and Cybernatics, vol. 30, pp. 249–278, 2001.
27. S. Volkwein, "Mesh-independence of an augmented Lagrangian-SQP method in Hilbert spaces and control problems for the Burgers equation," Diss. at the University of Technology Berlin, 1997.